

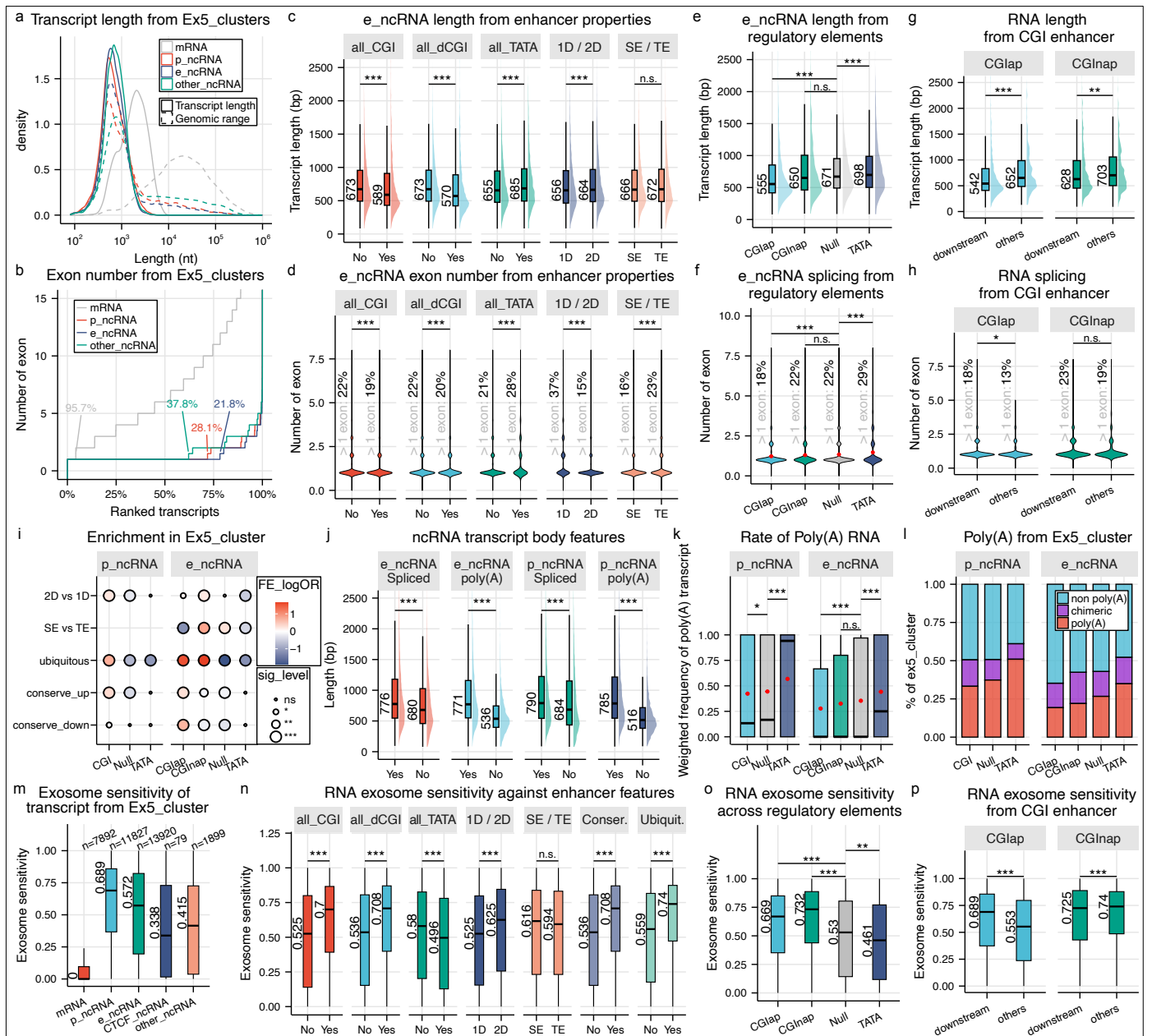


Figure 4 | Factors that affect the transcript model of eRNAs.

a, Transcript length and genomic range of RNA classes grouped by extended 5' end clusters (ex5_clusters). **b**, Median number of exons for each RNA class grouped by ex5_clusters. Percentages indicate groups with more than one exon. **c**, Comparison of e_ncRNA transcript lengths across genomic features: all_CGI (CpG island within ± 500 nt of summit), downstream CpG island (dCGI, within 500 nt downstream), TATA box (motif located -35 to -11 nt upstream), directionality (unidirectional, 1D; bidirectional, 2D), and enhancer type (super enhancer, SE; typical enhancer, TE) defined from iPSC, NSC or Neuron (excluding THP-1-derived enhancers). **d**, Median number of exons of e_ncRNAs grouped by genomic features. Percentages denote the fraction with more than one exon. **e**, Transcript length of e_ncRNAs grouped by mutually exclusive features; "Null" indicates ex5_clusters lacking any of the tested features. **f**, Median number of exons of the mutually exclusive e_ncRNA groups. **g**, Transcript length of e_ncRNAs comparing with and without the presence of dCGI, among CGIap and CGInap enhancers. **h**, Median number of exons comparing with and without the presence of dCGI. **i**, Enrichment of CGI, CGIap, CGInap, Null and TATA box across genomic features in e_ncRNA and p_ncRNA using Fisher's exact test (two-tailed). FE_logOR denotes the natural log of odds ratio. **j**, At the transcript level, presence of poly(A) and presence of splicing were evaluated for each model, and length distributions were compared across these features. **k**, Transcript models grouped by ex5_cluster with poly(A) frequency weighted by read count. Red dots indicate the mean values. **l**, Percentage of ex5_clusters produce RNA with poly(A) (weighted poly(A) frequency > 0.9), non-poly(A) (frequency < 0.1) and chimeric. **m**, Exosome sensitivity of transcript from Ex5_cluster, calculated using RNA-seq after EXOSC3 knockdown in iPSC, was grouped by ex5_cluster (weighted by read count). RNA categories include mRNA, p_ncRNA, e_ncRNA, CTCF_ncRNA, and other_ncRNA. **n** Exosome sensitivity of e_ncRNAs grouped by ex5_clusters and genomic features. Conservation was defined using 17-way phastCons scores from 500 nt upstream of the summit; clusters above the median score were classified as "Yes." Other feature definitions as in (c). **o**, Exosome sensitivity of enhancer-like ex5_clusters across mutually exclusive regulatory element groups. **p**, Exosome sensitivity of RNA derived from CGIap and CGInap enhancers, with and without dCGI. Unless specified, statistical tests were two-sided Wilcoxon; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.